

Low-fat dietary pattern and risk of benign proliferative breast disease: a randomized, controlled dietary modification trial



Modifiable factors, including diet, might alter breast cancer risk. We used the WHI Dietary Modification (DM) trial to test the effect of the intervention on risk of benign proliferative breast disease, a condition associated with increased risk of and considered to be on the pathway to invasive breast cancer. The WHI DM trial was a randomized, controlled, primary prevention trial conducted in 40 US clinical centers from 1993–2005. 48,835 postmenopausal women, aged 50–79 years, without prior breast cancer, were enrolled. Participants were randomly assigned to the DM intervention group or to the comparison group. The intervention was designed to reduce total dietary fat intake to 20% of total energy intake, and to increase fruit and vegetable intake to ≥ 5 servings/day and intake of grain products to ≥ 6 servings/day, but resulted in smaller, albeit significant changes in practice. Participants had biennial mammograms and regular clinical breast exams. We identified women who reported breast biopsies free of cancer, obtained the histologic sections, and subjected them to standardized central review. During follow-up (average, 7.7 years), 570 incident cases of benign proliferative breast disease were ascertained in the intervention group and 793 in the comparison group. The hazard ratio for the association between DM and benign proliferative breast disease was 1.09 (95%CI, 0.98–1.23). Risk varied by levels of baseline total vitamin D intake but it varied little by levels of other baseline variables. These results suggest that a modest reduction in fat intake and increase in fruit, vegetable, and grain intake does not alter the risk of benign proliferative breast disease.